Terms

Coexistence: The set of challenges posed by the sharing of the spectrum by multiple communication technologies.

Colocation: A situation when more than one radio technology is used inside the same device. Colocated radios can sometimes interfere with each other.

of time, the buffer will start to overflow and data will be lost.

Reliability challenges addressed by GFSK

The way in which radio signal properties are used for representing digital data is known as modulation scheme. The schemes based on frequency tend to be less susceptible to interference due to noise than those which are based on amplitude.

Bluetooth uses a special frequency based modulation scheme called Gaussian frequency shift keying (GFSK). It works by shifting a central carrier frequency upwards or downwards to represent either 1 or 0.

But abrupt changes in frequency can generate noise. Therefore, GFSK adds a filter, which causes frequency transitions to be smooth and less noisy. GFSK is a good fundamental platform on which to base your wireless communication.

Bluetooth packet fields

A Bluetooth packet has various fields. At the beginning is the preamble, which is 8-bit long and contains an alternating pattern of 1s and 0s. It is used for finding the precise transmitter frequencies and to set the radio's gain control.

Access address, which is 32-bit long, contains a special value, which means the packet can be received by any Bluetooth device or a unique identifier for a specific connection. Since a Bluetooth controller will be receiving both relevant and irrelevant signals, it must be able to distinguish and accurately pick out the relevant ones.

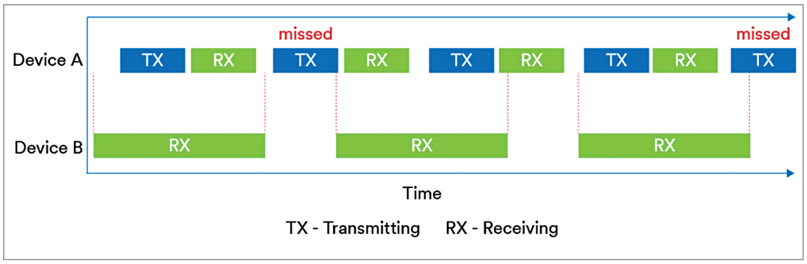

Figure 2: Missed packets in unsynchronised communication

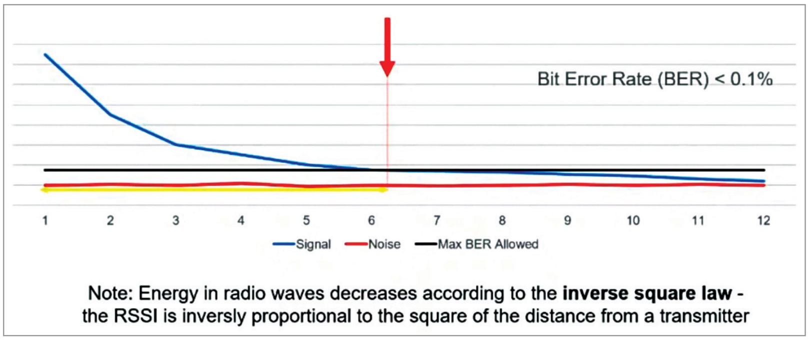

Figure 3: Energy decrease with distance

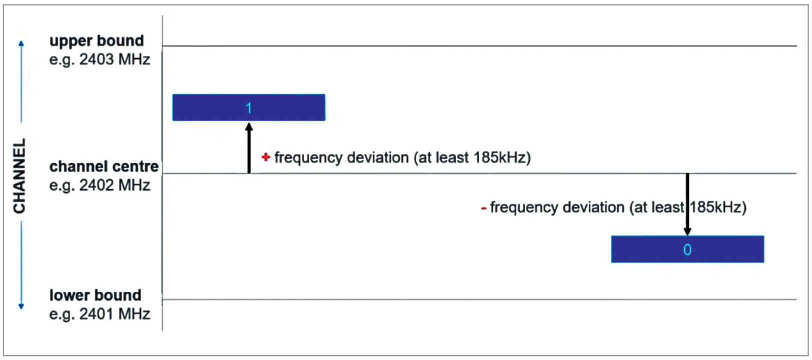

Figure 4: GFSK shifts a central carrier frequency upwards or downwards to represent 1 or 0

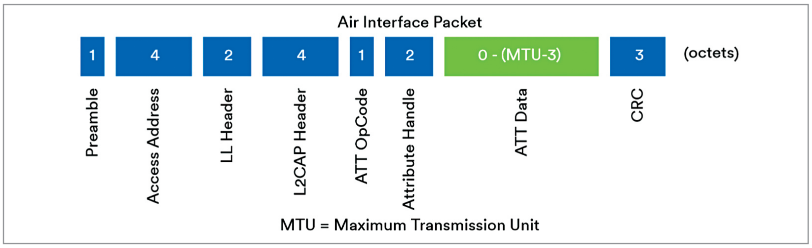

Figure 5: The fields in a Bluetooth packet

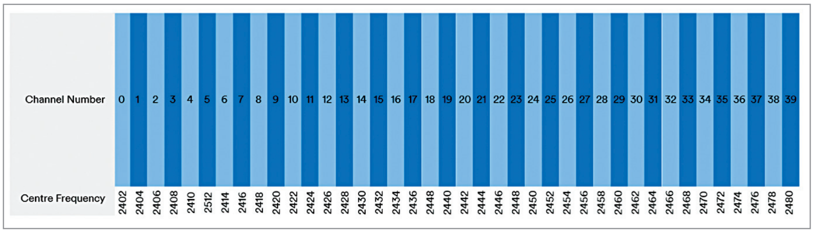

Figure 6: 2.4GHz band divided into 40 × 2MHz channels